

The empirical recurrence for OEIS A305514, recovered from its tail

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Abstract

OEIS A305514 records “Empirical recurrence of order 91” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 234$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 10$, so the recurrence is proved for every $n \geq 101$. Its last coefficient is $c_{91} = -4 \neq 0$, so no further tail satisfies anything shorter; any order-91 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A305514 is “Number of $n \times 6$ 0..1 arrays with every element unequal to 1, 2, 5, 6 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

5, 5, 19, 38, 108, 343, 925, 2627, ...

The entry states:

Empirical recurrence of order 91 (see link above)

As of the “Last modified” line on the live entry (Jun 03 2018 13:19:33, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 234$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 234$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 91: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 10$ is the first whose minimal order is exactly the stated 91.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 468$ exact terms from $n = 10$ on, it returns order 91 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{91} c_i a(n-i),$$

c_1	2	c_{24}	−11704	c_{47}	−28234	c_{70}	13663
c_2	3	c_{25}	2490	c_{48}	104442	c_{71}	6383
c_3	18	c_{26}	32889	c_{49}	−90853	c_{72}	3070
c_4	−43	c_{27}	−46752	c_{50}	14615	c_{73}	14281
c_5	−49	c_{28}	9517	c_{51}	71704	c_{74}	2033
c_6	−124	c_{29}	33829	c_{52}	−142071	c_{75}	−8220
c_7	344	c_{30}	−45447	c_{53}	14644	c_{76}	−4469
c_8	345	c_{31}	51738	c_{54}	125669	c_{77}	−182
c_9	287	c_{32}	−65385	c_{55}	−114079	c_{78}	−328
c_{10}	−1193	c_{33}	−17334	c_{56}	24262	c_{79}	238
c_{11}	−1221	c_{34}	118661	c_{57}	76940	c_{80}	783
c_{12}	659	c_{35}	−122692	c_{58}	−109176	c_{81}	413
c_{13}	513	c_{36}	12218	c_{59}	42181	c_{82}	574
c_{14}	2626	c_{37}	69892	c_{60}	47040	c_{83}	−196
c_{15}	−3900	c_{38}	−137661	c_{61}	−60915	c_{84}	−344
c_{16}	5389	c_{39}	112257	c_{62}	26796	c_{85}	246
c_{17}	−1824	c_{40}	20408	c_{63}	−14838	c_{86}	66
c_{18}	5006	c_{41}	−102147	c_{64}	−21327	c_{87}	−68
c_{19}	−17956	c_{42}	155853	c_{65}	14127	c_{88}	48
c_{20}	5617	c_{43}	−94368	c_{66}	−15841	c_{89}	26
c_{21}	4057	c_{44}	−74337	c_{67}	9957	c_{90}	−10
c_{22}	10179	c_{45}	176455	c_{68}	−10179	c_{91}	−4
c_{23}	−4157	c_{46}	−48129	c_{69}	−25152		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 101$, and it is the recurrence OEIS A305514 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 10$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{91} - c_1x^{91-1} - \dots - c_{91}$. Its constant term is $-c_{91} = 4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 91 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 91, so $\deg q = 91$, and it is monic. It holds from some index t on. From $\max(t, 10)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 91, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 91, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -4 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-91 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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